AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM358
Machine Learning &Artificial intelligence

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19/Nov./2025

Number of Exam Pages: ... (including this cover sheet)	**Time Allowed:** 120 minutes

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Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 ..... finds shallowest solution because always works on shallowest nodes first

A Best first
B BFS
C A*
D DFS

2 In the ..... search, given a state, we only know whether it is a goal state or not

A adaptive
B uninformed
C smart
D informed

3 ..... are the means by which robots move and change the shape of their bodies.

A GPS
B Effectors
C Sonar sensors
D proprioceptive sensors

4 ..... search from both the start and the goal, in parallel

A forward search
B Bidirectional search
C backward search
D None of the above

5 .....is a mechanism in which communication between a number of units results in a complex task being performed.

A interaction
B selection
C emergence
D adaptation

6 ..... means deciding the steps required to move blocks from an initial configuration (the start state) to another configuration (the goal state).

A planning
B Block world
C smart world
D None of the above

7 .....means "How long does it take to find a solution?"

A Optimality
B Time complexity
C Space complexity
D Completeness

8 ..... sensors measure range based on physical contact, and can be deployed only for sensing objects very close to the robot.

A Tactile sensors
B Range finders
C All of the above
D Sonar sensors

9 In the ..... search, given a state, we only know whether it is a goal state or not

A informed
B smart
C uninformed
D adaptive

10 .....can be defined as "it is the inability to see the consequences of moves"

A stochastic problem
B deterministic problem
C horizon problem
D None of the above

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

a. Explain how the mechanism of interaction contributes to the development of natural intelligence in species. [5 marks]

b. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	A	A	C	C	D	D	E	F	F	G	H	H	H	J	J
t	32	C	D	E	F	G	H	40	J	14	28	13	29	22	17	35	45

Given the aforementioned tree, use α-β pruning minimax algorithm to evaluate each node in the tree.

Question 3: (15 marks)

a. Explain how collaboration and communication are essential for achieving collective goals. Provide specific examples from the Argentine ants' collective behavior. [5 marks]

b. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	A	B	C	C	E	F	F	F	G	G	G	G
t	B	C	D	E	F	G	H	I	J	K	L	M	N	O

Given the aforementioned tree, apply DFS to traverse the tree showing all agenda content and traced paths until reaching the goal state if exist assuming that the goal state is L

Question 4: (15 marks)

a. Describe the process of planning using means-end analysis with STRIPS.[5 marks]

b. At a given point of time and given an ant, the ant found 5 possible routes at its journey. The pheromone amounts on each route are :

7.000000 8.000000 8.000000 3.000000 0.000000

If the length of each route which is inversely proportional to the route desirability is known to the ant and lengths are :

64.000000 7.000000 61.000000 67.000000 61.000000

Calculate the probability of selecting each route by that given ant

Assume Alpha=0.7 and beta= 0.6

End of Questions

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